

Teaching LLMs to Understand Arab Culture - All 22 Countries of It

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When LLMs gets Culture Wrong

When a large language model suggested that someone have a beer after prayer, it was not just a minor glitch, it was an example of cultural disconnect. Around 93% of the Arab world's population is Muslim, where, the consumption of alcohol contradicts religious practices and social norms. This recommendation revealed a fundamental problem: AI systems trained primarily on Western data often fail to grasp the cultural nuances of the communities they are meant to server.

This is not just about avoiding awkward suggestions. Artificial Intelligence has become more and more integrated into people's daily lives, meaning that the stakes for cultural sensitivity have never been higher. For the Arab world, home to over 450 million people across 22 countries, each with distinct traditions, dialects, and cultural practices, the need for culturally aware AI is critical.

Enter PALM Alwajih et al. [2025], a culturally inclusive and linguistically diverse dataset for Arabic LLMs, a groundbreaking project that spent an entire year building the first comprehensive Arabic instruction dataset that truly reflects the diversity of the Arab culture and language

The Arabic LLM Gap

The problem runs deeper than occasional cultural missteps. Most large language models are trained on data that's either originally in English or translated from English to Arabic. This approach introduces what researchers call "Anglocentric bias", where essentially the LLM thinks in English cultural terms, even when responding in Arabic.

Existing Arabic datasets have tried to address this, but they fall short in critical ways. In the paper, it includes some multilingual datasets like AYA which includes only 5000 Arabic instructions out of 204000 total entries. Others, like AraDICE, rely heavily on machine translation and data re-targeting rather

than original, culturally grounded content. CIDAR perhaps one of the most expansive datasets which contains 10000 Arabic instructions but it has limited geographic and dialectal diversity.

The Arab world is not monolithic. It spans two continents, vastly different climates from the Mediterranean coast to desert regions, and host cultures shaped by distinct historical experiences. Modern Standard Arabic (MSA) serves as the formal written language, but people speak in regional dialects that can differ as much as separate languages. Someone from Morocco might struggle to understand colloquial Yemeni Arabic without some adjustments.

Previous datasets rarely captured this complexity, focusing mainly on MSA and neglecting the rich tapestry of dialectal Arabic that people actually use on the daily bases.

Building PALM: A Community Effort

PALM took a radically different approach. Rather than relying on machine translation or a small team of annotators, the project assembled 44 researchers across the Arab world. This was not just about collecting data, it was about ensuring authentic representation.

The team succeeded in recruiting local experts from 15 of the 22 Arab countries. For the remaining 7 countries, they carefully selected annotators from neighboring regions with cultural familiarity. Within each country, team members came from different regions, ensuring that the dataset captured internal diversity as well.

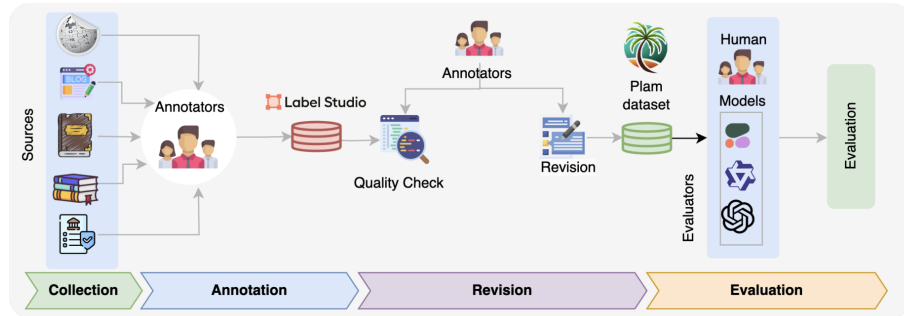


Figure 1: Pipeline for PALM creation

The annotation process was meticulous. There were around 100 pages of annotation guidelines, which emphasized using trustworthy sources such as academic books, and government websites, this was done to maintain objectivity and to avoid personal opinions, especially on sensitive topics like politics and

religion.

Results: A dataset of 17411 instructions pairs (input-output examples covering diverse topical areas. These range from local celebrations and traditional foods to geography, history, literature, and much more. Some topics are country specific, like Egyptian cuisine, while others span the Arab world in general, like science and technology.

Another thing to note is that PALM includes 4211 instructions pairs in dialectal Arabic, representing 10 different regional dialects. This makes it the most dialectically diverse Arabic instruction datasets created to date.

The dataset is split into 3 parts: a training set of 13559 instructions (publicly released for model development), a public test set of 1926 instructions (for benchmarking during development), and a private set of 1926 instructions (accessible only through a leaderboard to prevent data contamination).

Testing LLM’s Cultural IQ

Armed with PALM, the researchers set to evaluate how well current LLM models actually understand Arab culture. They tested 18 different LLMs, including both closed-source models like GPT-4o and Claude 3.5 Sonnet and open-source models such as LLaMA 3.1, Qwen 2.5, and Arabic-specific systems like JAIS and AceGPT.

The evaluation used multiple approaches. First, they examined surface-level attributes:

- Did the model respond in the same language as the prompt?
- Did it maintain the dialect of the instructions ?
- Did it generate repetitive text (a sign of poor performance)?

More substantially, they employed an "LLM-as-Judge" approach, using three strong models (GPT-4o, Command R+, and Qwen 2.5-72B) to evaluate responses on 4 criteria:

1. Correctness (factual accuracy)
2. Coherence (logical consistency)
3. Helpfulness (utility to the user)
4. Detail (depth and comprehensiveness)

Each generated response was scored on a scale of 1 to 10.

To validate this automated evaluation, they conducted human assessments on

a subset of responses. This correlation between human and automated judgments was strong, where a Pearson correlation of 0.76 and Intraclass Correlation Coefficient of 0.78, confirming that the automated approach was reliable.

The Results: Winners, Losers, and Surprises

GPT-4o and Claude 3.5 Sonnet emerged as top performers, achieving median correctness scores above 6.0. Command R+ and Qwen 2.5-72B followed closely (5.8-6.0), while smaller models like JAIS-13b and LLaMA-3.1-8B struggled significantly (3.0-4.0). Larger models with over 70 billion parameters consistently outperformed their smaller counterparts.

But the evaluation uncovered more revealing findings:

The Dialect Problem: Even when prompted with regional dialects, model responded in Modern Standard Arabic about 90% of the time. Dialectal consistency remained below 10% across all models. Current LLMs struggle to maintain the informal, colloquial language people actually use.

Geographic Disparities: Performance varied dramatically by country. GPT-4o scored 7.5 for Syrian content and 7.3 for Djibouti, but lower for others. Egypt and Tunisia consistently yielded higher scores, while Morocco proved more challenging. Some countries and dialects are clearly better represented in training data than others.

Topic Variations: LLMs performed better on celebrations, history, and travel, but struggled with sports and food, since reflecting the difference between standardized historical information and localized, experiential knowledge.

The Repetition Problem: Some models, particularly Gemma-2-9b, generated repetitive text in 42% of responses, showing a clear struggle with Arabic

Why This Matters

PALM’s findings reveal that even the “best” models have substantial room for improvement. Top performers scoring 6-7 out of 10 means they are getting roughly 30 - 40% of culturally specific information wrong. As for education, healthcare, or customer support applications, their error rate is problematic.

The near-complete absence of dialectal consistency represents a fundamental disconnect. When someone asks a question in Egyptian Arabic, they expect a response in Egyptian Arabic, not formal MSA. This is not just linguistic preference, dialects carry cultural markers and establish rapport.

The geographic disparities highlight an equity issue. If LLMs perform better on Egyptian content than Yemeni content, they risk reinforcing digital divides and

marginalizing underrepresented communities.

The good news is that PALM provides both a diagnostic tool and a remedy. By publicly releasing the training set (13,559 instructions), where researchers give developers concrete, culturally grounded data to improve their models.

Looking Forward

The researchers acknowledge PALM’s limitations. For 7 countries, they relied on annotators from neighboring regions rather than local speakers. Even for countries with local annotators, the dataset cannot capture every regional dialect. Automated evaluation may still misjudge subtle cultural elements.

Despite these limitations, PALM represents a significant step forward. It is the first dataset to systematically cover all 22 Arab countries with substantive culturally grounded content, which is entirely human-created and explicitly including dialectal diversity.

Future work might expand dialectal coverage, increase the representation of underrepresented countries, and develop more sophisticated evaluation methods. The project demonstrates the work of community-driven approaches: involving local experts ensures authenticity that no machine translation could achieve.

A New Standard

PALM sets a new standard for culturally aware language technologies. The dataset is fully open-source (PALM Dataset), inviting the broader research community to build upon their work.

For 450 million Arabic speakers across 22 countries, PALM offers a glimpse of hope that LLMs can move beyond Western-centric assumptions. The "beer after prayer" mistake should become a relic of the past, not just because we have hidden the problem, but because we have done the hard work of teaching AI about the cultures it serves.

Therefore, building inclusive LLMs requires including the communities it aims to serve, not just in testing but also in creation. PALM proves that it can be done.

References

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